INTRODUCTION TO COMPUTER GRAPHICS

Course Objective

- To make students familiar with the basic steps of the graphics pipeline.
- To make students understand the theories and underlying mathematics of graphics applications.
- To understand the algorithms commonly used in 3D computer graphics.
- To make the students understand advanced computer graphics techniques and applications.
- To demonstrate how to solve problems by designing projects.

Course Outcomes (COs)

- Demonstrate advanced knowledge of the fundamentals of 2D and 3D computer graphics.
- Explain and apply the algorithms commonly used in 3D computer graphics.
- Display competency in a number of advanced computer graphics techniques and applications.

Text and Reference Books:

Sl.	Title	Author(s)	Publication Year	Edition	Publisher	ISBN
01	Computer Graphics: Principles and Practice	John F. Hughes, James D. Foley, Andries van Dam, Steven K. Feiner	1982	**2nd**	Pearson	ISBN-13: 978-0201848403 ISBN-10: 9780201848403

Assessment Tools:

Assessment Tools	**Weightage (%)**
Assignments	5%
Quizzes (Avg. of best n-1)	15%
Laboratory Work	25%
Midterm Examination	20%
Final Examination	35%

Topics

- Basics of Computer Graphics
- Scan Conversion Algorithms: DDA, Midpoint Line Drawing Algorithm
- Eight Way Symmetry
- Line Clipping Algorithms: Cohen-Sutherland, Cyrus-Beck
- Geometric Transformations
- Projection
- Color Model
- Lighting Model
- Shading Models
- Curves

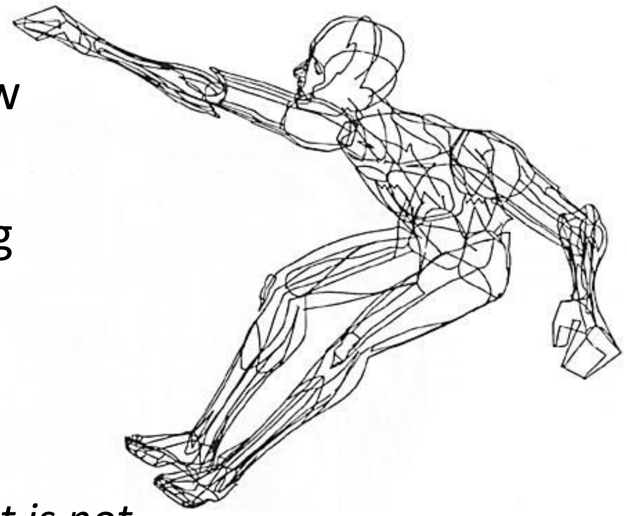
What is Computer Graphics?

- Computer graphics generally means creation, storage and manipulation of models and images
- Such models come from diverse and expanding set of fields including physical, biological, mathematical, artistic, and conceptual/abstract structures

Frame from animation by William Latham, shown at **SIGGRAPH 1992**. Latham creates his artwork using rules that govern patterns of natural forms.

What is Computer Graphics?

- William Fetter coined term “computer graphics” in 1960 to describe new design methods he was pursuing at Boeing for cockpit ergonomics
- Created a series of widely reproduced images on “pen plotter” exploring cockpit design, using 3D model of human body.

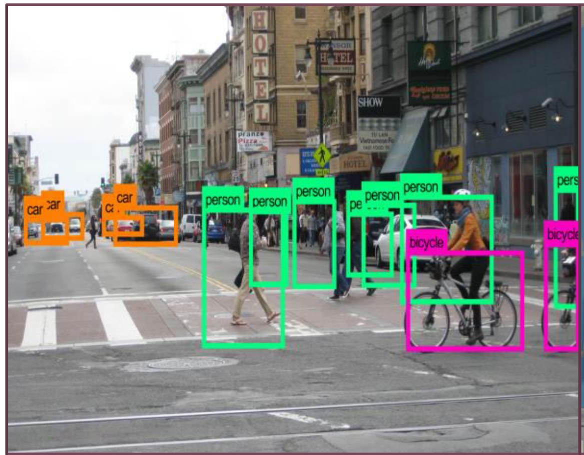


“Perhaps the best way to define computer graphics is to find out what it is not. It is not a machine. It is not a computer, nor a group of computer programs. It is not the know-how of a graphic designer, a programmer, a writer, a motion picture specialist, or a reproduction specialist.

Computer graphics is all these – a consciously managed and documented technology directed toward communicating information accurately and descriptively.”

Computer Graphics, by William A. Fetter, 1966

Differences



Computer vision: **Extracting information** from the contents of an input image or video frame.

Ex: Face Recognition, Autonomous Driving

Computer graphics: **Creating** an image **from scratch** using computer.

Ex: Animated Movies

Digital image processing: **Processing raw input images** to perform different operations.

Ex: Apply Filter on an Image, Noise Reduction, Compression

Applications of Computer Graphics

1. **Entertainment & Gaming:**

   Visual Effects, Animation, Video Games, VR/AR Experiences.

2. **Design & Engineering:**

   CAD Modeling, Architectural Visualization, Product prototyping.

3. **Education & Training:**

   Virtual Labs, Interactive simulations, VR-based Learning.

4. **Medical & Scientific Visualization:**

   MRI/CT Imaging, Surgery Simulation, Data and Astrophysical Visualization.

5. **User Interfaces & Marketing:**

   GUIs, Graphic Design, 3D Product Visualization, Digital Ads.

Input Methods in Computer Graphics

- **Keyboard and Mouse**: Traditional devices for controlling software, navigation, and object manipulation.
- **Graphics Tablets**: Used for digital drawing and illustration.
- **Touchscreens**: For direct interaction with graphical elements.
- **3D Scanners**: Capture the 3D structure of objects to create digital models.
- **Body as Interaction Device**:

Xbox Kinect

Leap Motion

Nimble UX

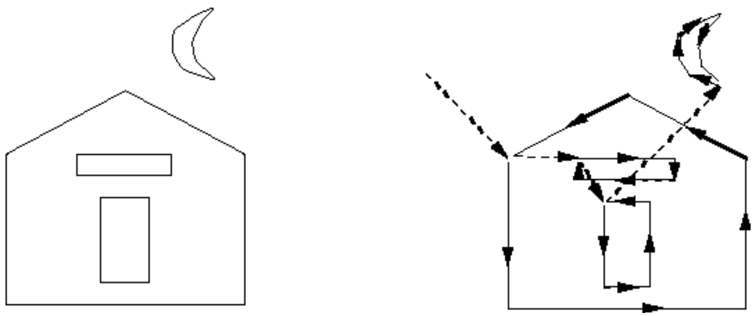
Output Devices in Computer Graphics

- **Monitors and Screens**: The most common output devices, used to display images and videos.
- **Printers**: For generating hard copies of graphics (2D or 3D).
- **Projectors**: Used in large-scale displays for presentations and entertainment.
- **VR/AR Headsets**: Immersive displays for virtual and augmented reality applications.

Graphics Display Hardware

Vector Display (calligraphic, stroke, random-scan)

- Driven by display commands
  - (move (x, y), char("A") , line(x, y)…)
- Survives as "scalable vector graphics"

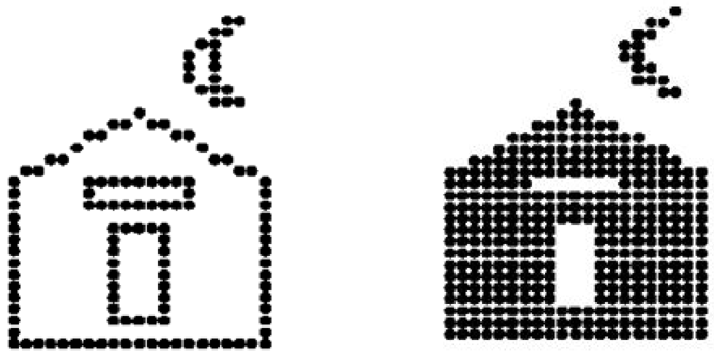


Ideal Drawing

Vector Drawing

Raster Display (TV, bitmap, pixmap) used in displays and laser printers

- Driven by array of pixels (no semantics, lowest form of representation)
- Note "jaggies" (aliasing errors) due to discrete sampling of continuous primitives



Outline

Filled

Output Technology: Vector Displays

- Also called Calligraphic, Stroke or Line Drawing Graphics
- Lines drawn directly on phosphor
  - Display processor directs electron beam according to list of lines defined in a "**display list**"
  - Phosphors glow for only a few microseconds so lines must be redrawn or **refreshed** constantly
  - Deflection speed limits # of lines that can be drawn without flicker.

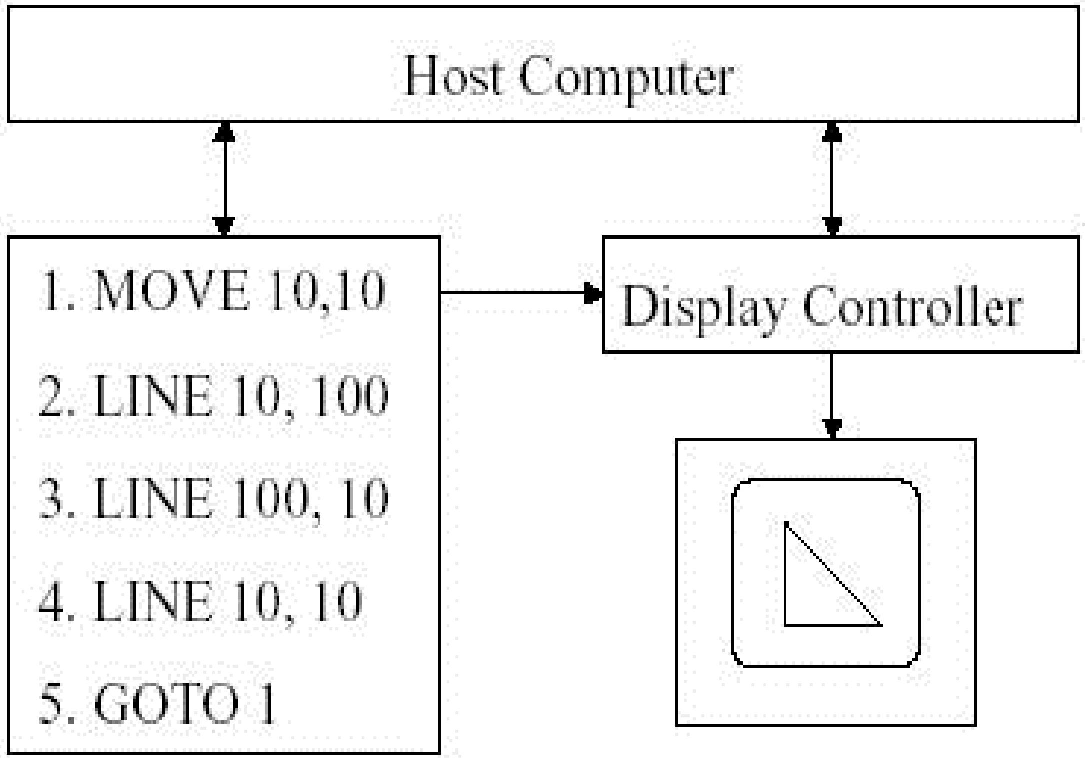


Fig.: Vector Display Architecture

Output Technology: Raster Displays

- Display **primitives** (lines, shaded regions, characters) stored as pixels in **refresh buffer** (or **frame buffer**)
- Electron beam scans a regular pattern of horizontal raster lines connected by horizontal retrace and vertical retrace
- Video controller coordinates the repeated scanning
- Pixels are individual dots on a raster line

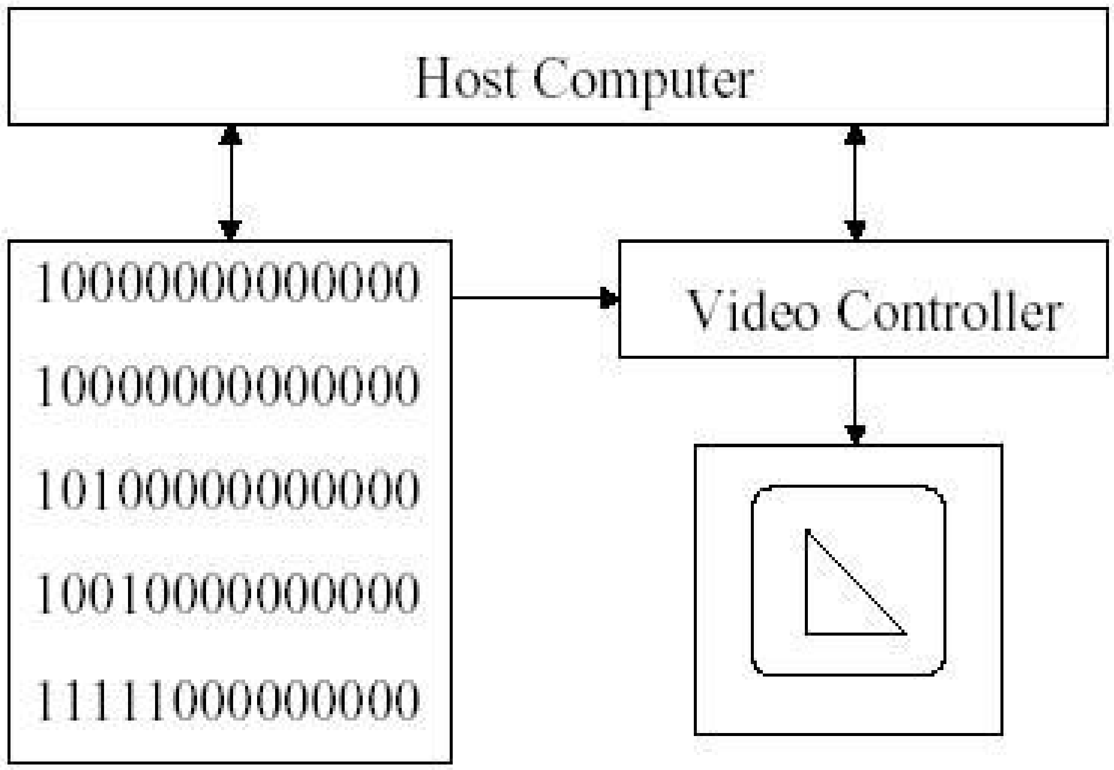
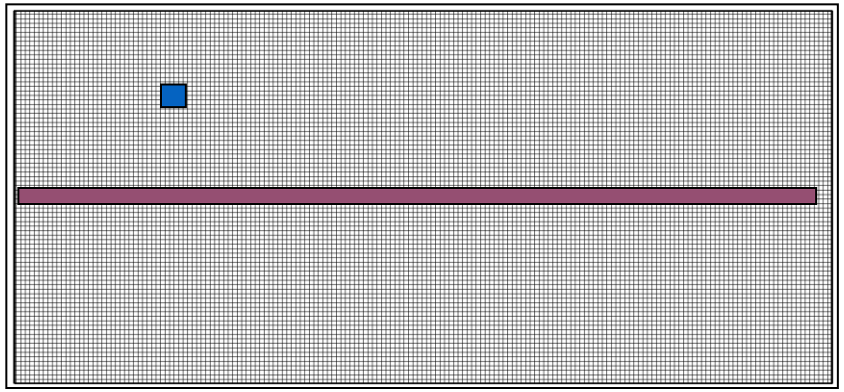


Fig.: Raster Display Architecture

Basic Definitions

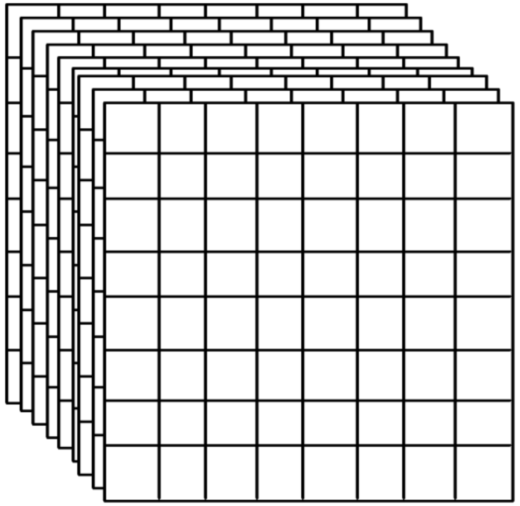
- **Raster:** A rectangular array of points or dots.
- **Pixel (Pel):** One dot or picture element of the raster
- **Pixel Grid:** Images are arranged in a grid of pixels, where each pixel is stored with specific color information (RGB values)
- **Scan line:** A row of pixels



Video raster devices display an image by sequentially drawing out the pixels of the scan lines that form the raster.

Frame Buffer

- A frame buffer is characterized by its size, x, y, and pixel depth.
- the **resolution** of a frame buffer is the number of pixels in the display. e.g. 1024x1024 pixels.
- Bit Planes or Bit Depth is the number of bits corresponding to each pixel. This determines the **color resolution** of the buffer.
- Dual ported (simultaneously writing values and displaying in the monitor)



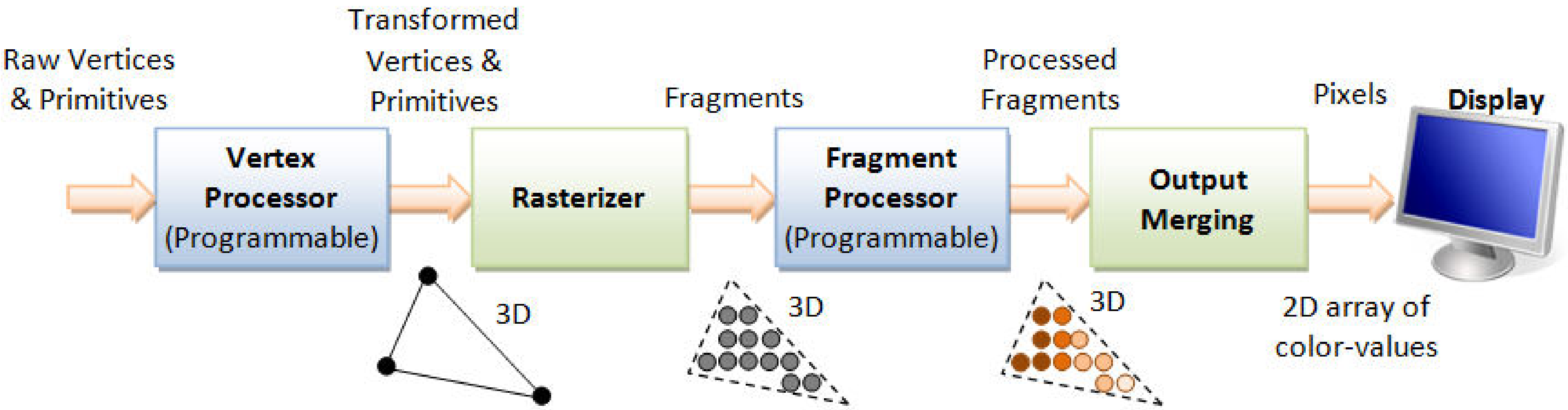
Bilevel or monochrome displays have 1 bit/pixel (128Kbytes of RAM)
8bits/pixel -> 256 simultaneous colors
24bits/pixel -> 16 million simultaneous colors

Raster vs. Vector Displays

- **Raster**: Pixel-based, fixed resolution, common in displays. Cheaper. More realistic.
- **Vector**: Shape-based, resolution-independent, best for scalable designs.
- **Use Cases**: Raster is used for photographs and complex images, while vector is used for fonts and logos.

Rendering Pipeline

The rendering pipeline is a sequence of steps used in computer graphics to convert a 3D scene into a 2D image on the screen.

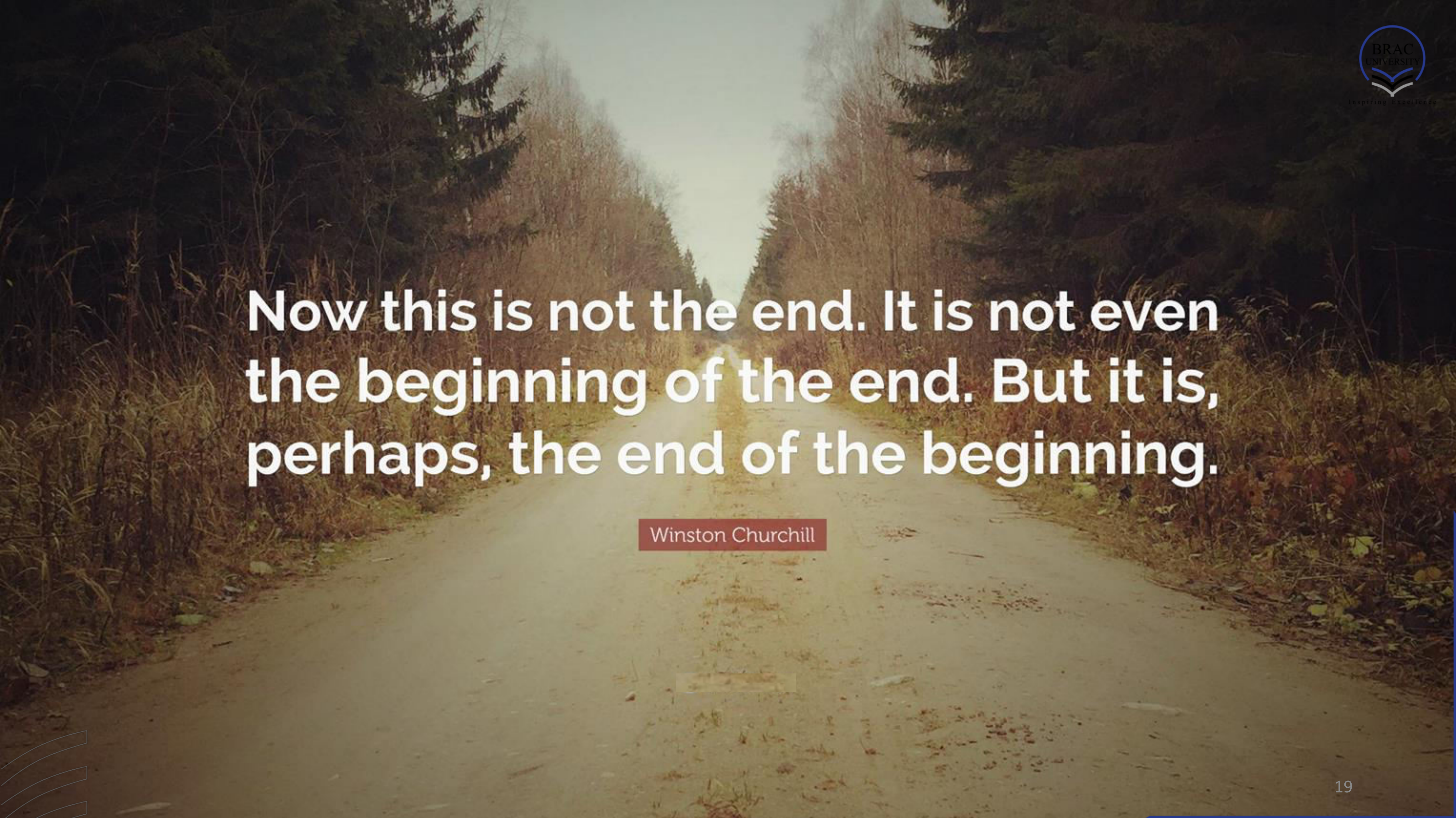


3D Graphics Rendering Pipeline: Output of one stage is fed as input of the next stage. A vertex has attributes such as (x, y, z) position, color (RGB or RGBA), vertex-normal (n_x, n_y, n_z), and texture. A primitive is made up of one or more vertices. The rasterizer raster-scans each primitive to produce a set of grid-aligned fragments, by interpolating the vertices.

Rendering Pipeline

The 3D graphics rendering pipeline consists of the following main stages:

1. **Vertex Processing**: Process and transform individual vertices.
2. **Rasterization**: Convert each primitive (connected vertices) into a set of fragments. A fragment can be treated as a pixel in 3D spaces, which is aligned with the pixel grid, with attributes such as position, color, normal and texture.
3. **Fragment Processing**: Process individual fragments.
4. **Output Merging**: Combine the fragments of all primitives (in 3D space) into 2D color-pixel for the display.


Now this is not the end. It is not even the beginning of the end. But it is, perhaps, the end of the beginning.
Winston Churchill